

Estradiol

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This estradiol assay is designed for the investigation of fertility of women of reproductive age and for the support of in vitro fertilization.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

E2 levels in children, postmenopausal women, and men are much lower than in women of reproductive age. The increased sensitivity and specificity that are achieved by LC/MS-MS are the more appropriate choice for these clinical situations than the electrochemiluminescence immunoassay (ECLIA) method.^{1,2} See [Estradiol, Sensitive, LC/MS \[140244\] \(/tests/140244/estradiol-sensitive-lc-ms\)](#). LC/MS-MS offers superior analytical sensitivity, specificity and a larger dynamic range than immunoassays.¹ The clinical applications benefiting from highly sensitive E2 measurement include the assessment of congenital defects in sex steroid metabolism and disorders of puberty. This sensitive assay also has application in the evaluation of estrogen deficiency in men and menopausal women, fracture risk assessment in these populations, and increasingly, in therapeutic drug monitoring of low-dose female hormone replacement therapy or antiestrogen treatment.

Estradiol levels tend to fluctuate dramatically during the perimenopausal transition. There is significant overlap of the expected range in menopausal women with values observed during normal menstrual cycles. Estradiol results obtained with different assay methods cannot be used interchangeably in serial testing. To monitor a patient's serial results, it is best to ensure that the same methodology is used each time the test is performed.

As with all test containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes. In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur. The test contains additives, which minimize these effects.

Due to the risk of cross-reactivity, this estradiol assay should not be used when monitoring estradiol levels in patients treated with fulvestrant (Faslodex®). An alternate method that is not subject to interference by this drug such as the Estradiol, Sensitive, LC/MS [140244] (/tests/140244/estradiol-sensitive-lc-ms) should be used to measure estradiol levels in patients treated with fulvestrant.

Custom Additional Information

Estradiol is responsible for the regulation of the estrous and menstrual female reproductive cycles and for the development and maintenance of female secondary sex characteristics.^{3,4} Estradiol plays a key role in germ cell maturation and numerous other, non–gender-specific processes, including growth, bone metabolism, nervous system maturation, and endothelial responsiveness. Estrogens are crucial for the normal development and maintenance of the breasts and the uterus.⁵ However, excessive estrogen levels can promote cell proliferation and may increase the risk of developing breast and uterine cancer as well as uterine endometriosis.⁵

The three major naturally occurring estrogens in women are estrone (E1), estradiol (E2), and estriol (E3). E2 is the predominant estrogen during reproductive years, both in terms of absolute serum levels as well as in terms of estrogenic activity.³ During menopause, a dramatic drop in E2 production leaves estrone as the predominant circulating estrogen. Estriol is the main pregnancy estrogen, but it does not play a significant role in nonpregnant women or men.³ The concentration of E2 in men is much lower than in women of reproductive age. All estrogens are synthesized from androgen precursors by the enzyme aromatase.^{3,5} Aromatase converts the androgenic substrates androstenedione, testosterone, and 16-hydroxytestosterone to the corresponding estrogens: estrone, estradiol, and estriol.⁵ E2 is produced primarily in ovaries and testes by aromatization of testosterone.³ A lesser amount of E2 is produced in the adrenal glands and some peripheral sites, most notably adipose tissue. Most of the circulating estrone is derived from peripheral aromatization of androstenedione (mainly in the adrenal gland). E2 and E1 can be converted to each other, and both are inactivated via hydroxylation and conjugation. E2 demonstrates two to five times the biological potency of E1.³

The importance of E2 testing and the need for reliable and accurate estradiol measurements throughout the analytic range are emphasized in several recent publications.⁶⁻⁸ LabCorp offers a sensitive estradiol by LC/MS (**140244**) (</tests/140244/estradiol-sensitive-lc-ms>). Measurement of serum E2 serves an integral role in the assessment of reproductive function in females and in the assessment of infertility, oligomenorrhea, and menopausal status. E2 is commonly measured for monitoring ovulation induction, as well as during preparation for in vitro fertilization. Because of the relatively high serum concentrations of E2 in these patients, readily available automated immunoassay methods with modest sensitivity meet the clinical requirements.

Adult female. In premenopausal women, E2 levels, along with luteinizing hormone (LH) and follicle-stimulating hormone (FSH) levels, delineate the stage of the menstrual cycle.³ E2 levels are lowest during the early follicular phase and rise gradually. Two to three days before ovulation, estradiol levels start to increase much more rapidly to a peak just before ovulation. This dramatic increase in circulating E2 levels induces a surge in LH and FSH. E2 levels decline modestly during the ovulatory phase and then increase again gradually until the midpoint of the luteal phase and ultimately decline back to early follicular levels.

Assessment of E2 levels is useful for the evaluation of hypogonadism and oligomenorrhea in women. Decreased ovarian estrogen production is classified as hypergonadotropic or hypogonadotropic, depending on whether the disease is of gonadal or pituitary/hypothalamic origin.⁹⁻¹¹ Measurement of gonadotropins (LH and FSH) is fundamental in differentiating these two low estradiol states. The main causes of primary gonadal failure (hypergonadotropic) are genetic (Turner syndrome, familial premature ovarian failure), autoimmune (autoimmune ovarian failure, autoimmune polyglandular endocrine failure syndrome type II), and toxic (related to chemotherapy or radiation therapy for malignant disease).

Low E2 with low or inappropriately "normal" LH and/or FSH in young adult females is consistent with hypogonadotrophic hypogonadism.¹¹⁻¹³ This can be caused by hypothalamic or pituitary failure due to conditions including multiple pituitary hormone deficiency and Kallmann syndrome. Diagnostic workup includes the measurement of E2, along with pituitary gonadotropins and prolactin and, possibly, imaging. This endocrine presentation can be caused by starvation, overexercise, severe physical or emotional stress, and drug/alcohol abuse. While early studies suggested that E2 levels could be used to predict ovarian reserve in women of reproductive age undergoing assisted reproduction procedures, more recent studies have found the marker less useful.¹⁴ Estradiol measurement is useful in assessing the status of ovulation induction in women with hypogonadotropic hypogonadism,¹⁵ and for the prediction and prevention of ovarian hyperstimulation syndrome in patients undergoing assisted reproduction.¹⁶

Normal or high E2 with irregular or absent menstrual periods is suggestive of possible polycystic ovarian syndrome, androgen producing tumors, or estrogen producing tumors. In these cases, measurement of total and bioavailable androstenedione, dehydroepiandrosterone (sulfate), and sex hormone-binding globulin can aid in differential diagnosis.

The main site of estrogen biosynthesis in the nonpregnant premenopausal woman is the ovarian granulosa cells; however, the adipose tissue becomes a major source of circulating estradiol in postmenopausal women.³ After menopause, androstenedione, secreted by the adrenal gland, is converted into estrone in the adipose tissue.³ The conversion of plasma androstenedione to estrone increases with excess body weight in both pre- and postmenopausal women.³ Estrone is then eventually converted to estradiol by 17- β -hydroxysteroid dehydrogenase enzymes present in peripheral tissues.³

Measurement of E2 level, together with FSH and/or anti-Müllerian hormone (AMH) can be useful in predicting the timing of the transition into menopause.^{17,18} A large population study (Randolph) found that the mean E2 level started to decline approximately two years prior to the final menstrual period (FMP) and exhibited a maximal rate of change at the FMP. The mean E2 level stabilized a menopausal level approximately two years after FMP.¹⁷ A sensitive estradiol assay is required to measure E2 levels accurately in postmenopausal women. The current recommendations for postmenopausal female hormone replacement are to administer therapy in the smallest beneficial doses for as briefly as possible. Estrogen replacement in reproductive-age women should aim to mimic natural estrogen levels as closely as possible, while levels in menopausal women should be held near the lower limit of the premenopausal female reference range. Postmenopausal women with lower E2 levels are at increased risk of osteoporotic fractures, while higher estradiol levels are associated with increased risk of malignancy and cardiovascular disease.^{19,20} Accurate measurement of E2 in women receiving hormone replacement may play a role in optimizing therapy.

Gonadotropin receptor hormone (GNRH) analogues are used therapeutically to reduce the ovarian production of estradiol in sex hormone-dependent disorders, including endometriosis and uterine fibroids.²¹ Aromatase inhibitors are also used therapeutically to reduce circulating estrogens (E2 and E1) levels in hyperestrogenic conditions (ie, endometriosis in women and gynecomastia in men) and in estrogen-sensitive malignancies.²²⁻²⁶ The complete or near complete suppression of estradiol production induced by these treatments produces low serum levels that can only be accurately measured by sensitive methods.²⁷ See **Estradiol, Sensitive (LC/MS) [140244]** (</tests/140244/estradiol-sensitive-lc-ms>).

Adult male. The use of a sensitive, LC/MS assay for serum E2 measurement in males is preferred over direct immunoassays because of its greater sensitivity and lesser interference by other steroids.²⁸ See LabCorp test **Estradiol, Sensitive (LC/MS) [140244]** (/tests/140244/estradiol-sensitive-lc-ms). In males, estradiol is present at low concentrations in blood, but it is extraordinarily high in semen.³ Estradiol plays an important role in epididymal function and sperm maturation and is essential for normal spermatogenesis and sperm motility.³

Gynecomastia refers to a syndrome of abnormal feminization with swelling of the breast tissue in boys or men, caused by an imbalance of the hormones estrogen and testosterone.²⁹ Gynecomastia is common during puberty in boys and can be seen in older males due to increased estrogen levels related to obesity (increased aromatase activity), decreased hepatic clearance, estrogen ingestion, and estrogen-producing tumors. Asymptomatic gynecomastia is common in older men, but individuals who present with gynecomastia of recent onset (associated with pain and tenderness) may require clinical workup.²⁹ Gynecomastia and other signs of male feminization may be caused by an absolute increase in E2 and/or E1. The testes may directly secrete too much estradiol due to a Leydig-cell or Sertoli-cell tumor. They may also secrete estradiol indirectly through the stimulatory effect of a human chorionic gonadotropin-secreting tumor of gonadal or extragonadal germ-cell origin.²⁹ Alternatively, men with normal estrogen levels can develop gynecomastia, if testosterone levels are low due to primary/secondary testicular failure, resulting in an abnormally elevated estrogen:androgen ratio. Feminization may also occur in men treated with antiandrogen therapy or drugs with antiandrogenic effects (eg, spironolactone, digitalis). Conversely, individuals with elevated androgen levels will often exhibit gynecomastia caused by aromatase-catalyzed estrogen production.

Estrogens (and androgens) play an important role in the normal physiology of the skeleton in both sexes.³ Males with diminished estrogen levels (due to congenital aromatase deficiency) or insensitivity to estrogens (due to estrogen receptor deficiency) have a characteristic phenotype with regard to bone development.^{3,25} These males exhibit significant increased overall height due to lack of estrogen-induced epiphyseal closure.²⁵ The importance of estradiol in bone health is further supported by the fact that estradiol levels correlate better with bone mineral density than do testosterone levels in aging men.²⁵ The Endocrine Society has recently reported that low estradiol levels are associated with increased fracture risk and accelerated bone loss in older men.³⁰

Children and adolescents. A sensitive method is required to measure accurately the E2 concentrations found in boys and prepubertal girls. See LabCorp test **Estradiol, Sensitive (LC/MS) [140244]** (/tests/140244/estradiol-sensitive-lc-ms). Levels in boys and heavier girls are generally lower than in girls of normal weight.^{31,32} Adrenal steroids tend to increase prior to gonadal steroids at the beginning of the pubertal transition.³¹ In girls, E2 concentrations increase just before breast development.³¹

In precocious puberty (PP), estradiol and the gonadotropins, LH and FSH, tend to be above the prepubertal range.³³ E2 measurement in children suspected of having PP is performed to support the diagnosis and to determine the origin of the condition or disease. The source of increased estradiol can be exogenous estrogens or an ovarian cyst that has produced transient estrogens. Elevation of E1 or E2 alone suggests pseudoprecocious puberty, possibly due to a steroid-producing tumor.

It is not normal for an adolescent to be amenorrheic for greater than three months, even in the early gynecologic years,³⁴ and menstrual cycle duration persistently outside 21 to 45 days in adolescents is unusual.³⁵ Since estrogen deficiency is a risk factor for later development of osteoporosis and cardiovascular disease, a workup including sensitive E2 measurement is recommended for adolescent girls and women with potentially disordered hypothalamic-pituitary-gonadal function.^{11,34} Persistently low estrogens and elevated gonadotropins in children with delayed puberty suggest primary ovarian failure, while low gonadotropins suggest hypogonadotrophic hypogonadism. In this latter case, Kallmann syndrome (or related disorders) or hypothalamic/pituitary tumors should be excluded in well-nourished children.³⁶ Both E2 and E1 levels are very low or undetectable in children with aromatase deficiency.³⁵ Affected girls have hypergonadotropic hypogonadism, fail to develop secondary sexual characteristics, and exhibit progressive virilization.³⁵ The affected boys exhibit normal male sexual differentiation and pubertal maturation. However, boys with aromatase deficiency are typically extremely tall with eunuchoid proportions and continued linear growth into adulthood, severely delayed epiphyseal closure, and osteoporosis due to estrogen deficiency. Highly sensitive E2 measurement can be of value in the assessment of therapeutic efficacy of estrogen replacement in hypogonadal girls.³²

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

- Adult male: 7.6–42.6 pg/mL
- Adult female:
 - Follicular: 12.5–166.0 pg/mL
 - Ovulation: 85.8–498.0 pg/mL
 - Luteal: 43.8–211.0 pg/mL

- Postmenopausal: <6.0–54.7 pg/mL
- Pregnancy: First trimester: 215.0 to >4300.0 pg/mL
- Children (1 to 10 years):
- Male: 0.0–20.0 pg/mL
- Female: 6.0–27.0 pg/mL

Note: The results displayed above were obtained with the Roche ECLIA methodology.

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

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Footnotes

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